Predicting Risk in Shelter Animals

Course Project - Advanced Topics in Machine Learning
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Section 1: Introduction

1.1 Problem Description and Significance

Animal shelters across the United States face a constant challenge of overpopulation and limited resources. The ability to make data-driven decisions at the moment of intake is crucial for operational efficiency and animal welfare. This project utilizes the **"Louisville Metro KY - Animal Service Intake and Outcome"** dataset to address a critical classification problem: identifying animals at high risk of negative outcomes (**Euthanasia, Death, or Disposal**) immediately upon their arrival.

Unlike standard inventory or commercial prediction problems, the cost of error in this domain is ethical and irreversible. A **"False Negative"** (predicting an animal is safe when it is actually at risk) leads to missed opportunities for medical or behavioral intervention, potentially resulting in preventable euthanasia. To address this high-stakes environment, our research evaluates a diverse suite of predictive architectures-ranging from traditional linear models to state-of-the-art gradient boosting machines-to ensure the highest possible PR AUC. This project serves as a decision-support system designed to prioritize attention for the most vulnerable animals, shifting shelter management from reactive to proactive care.

1.2 Objectives

The primary goal of this study is to provide shelter staff with a robust predictive tool that flags high-risk animals with high sensitivity. The specific objectives are:

- **System Architecture:** Constructing an end-to-end automated pipeline for translating raw, noisy administrative logs into structured, high-quality data suitable for machine learning.

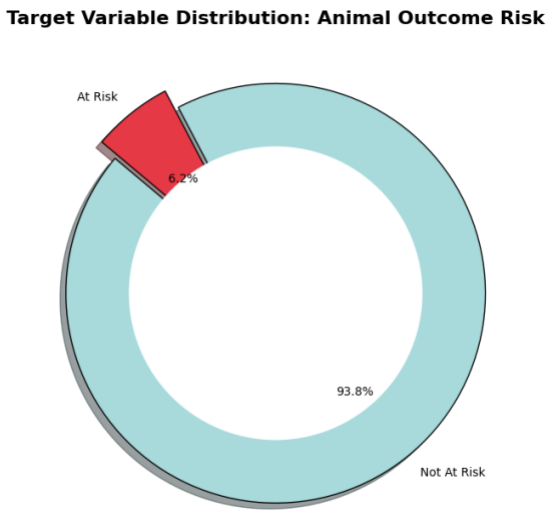
- **Model Benchmarking & Optimization:** We conducted an extensive comparative analysis of five distinct architectures: Logistic Regression, Random Forest, Multi-Layer Perceptron (MLP), CatBoost, and LightGBM. The primary objective was to identify the optimal architecture for categorical tabular data. We utilized hyperparameter tuning via **RandomizedSearchCV** to maximize **PR AUC**, which serves as the most reliable performance indicator for imbalanced classification. This approach ensures a robust trade-off between sensitivity and precision, allowing us to identify at-risk animals with a high degree of confidence while effectively minimizing the noise of false alarms.
- **Handling Class Imbalance:** Implementing specialized techniques, such as cost-sensitive learning (class weighting) and SMOTE, to prevent model bias toward the majority "Safe" class and ensure that at-risk animals are accurately detected.
- **Risk Profiling & Clustering Validation:** Utilizing unsupervised learning to uncover latent "Risk Archetypes". This objective includes a comparative analysis between **K-Modes clustering** (optimized for categorical variables) and **Hierarchical Clustering**, using validation metrics such as the **Silhouette Score**, WSS, and BSS to identify the most cohesive risk profiles. The analysis aims to understand the interaction between seasonality, pet size, and intake conditions to provide actionable insights for shelter operations.

Section 2: Dataset and Features

2.1 Dataset Overview

The raw dataset consists of over 64,000 records of animal intakes from the **Louisville Metro Animal Services** (sourced via Data.gov). Each record captures a snapshot of an animal's entry into the system, containing static attributes (Breed, Color, Sex) and dynamic situational attributes (Intake Type, Condition, Intake Date, and Kennel Location).

Target Definition: We derived a binary target variable, is_at_risk, to reflect the ethical priorities of shelter management. Outcomes such as **Euthanasia, Died, Disposal, and Hospice** were aggregated into the positive class (1), representing cases where intervention was either unsuccessful or required. Conversely, **Adoption, Return to Owner, Transfer, and Foster** were mapped to the negative class (0), representing successful shelter exits.



2.2 Data Cleaning and Preprocessing

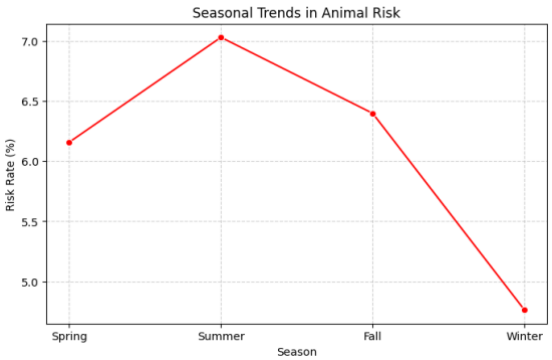
To transform administrative logs into a reliable predictive format, several preprocessing steps were taken, guided by the need to avoid data leakage and noise:

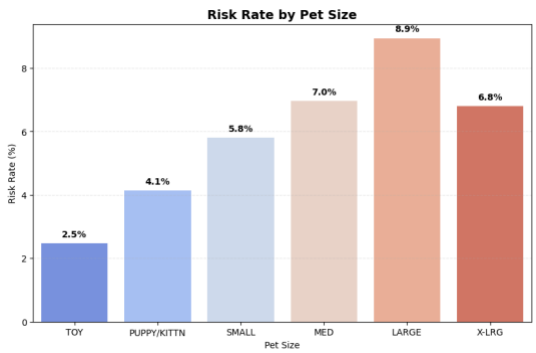
- **Noise Reduction & Leakage Prevention:** We removed records labeled as "RELOCATE" or "RELEASE" (primarily wildlife) and, crucially, filtered out animals that were "Dead on Arrival" (DOA). **Rationale:** Including DOA animals would create a "shortcut" for the model, as no shelter intervention could have altered their outcome, thus inflating performance metrics without providing clinical value.
- **Categorical Unification:** Features like IntakeCondition contained redundant values (e.g., "AGED" vs. "GERIATRIC"). We mapped these to unified categories to reduce sparsity and improve model convergence.
- **Missing Values:** Rows with missing critical identifiers or outcome types were removed. We chose not to use imputation for the target-related fields to avoid introducing artificial bias into our ground-truth labels.

2.3 Feature Engineering

Feature engineering was a pivot point in this project, where we translated domain knowledge into predictive power:

- **Pet Size Consolidation (New):** The raw data contained granular age/size markers. We consolidated these into six primary categories: PUPPY/KITTN, TOY, SMALL, MED, LARGE, and X-LRG. **Rationale:** Our EDA revealed that the PUPPY/KITTN group faces unique medical vulnerabilities, making this a high-impact feature for identifying risk.
- **Temporal Seasonality:** We extracted Month and Season from the intake timestamp. This captures "Kitten Season" trends (Summer), where the surge in intake often correlates with higher mortality rates due to resource strain.
- **Breed Complexity & Cleaning:** The Breed column originally contained hundreds of unique strings. We implemented text parsing to extract a binary mixed_breed feature and standardized breed names. This addresses the high-cardinality problem, allowing models like LightGBM to find patterns without overfitting on rare breeds.
- **Recidivism (*first_time*):** By analyzing unique AnimalIDs, we determined if an intake was a first-time entry or a return. **Rationale:** Returning animals may indicate underlying behavioral or chronic health issues that were not resolved during their first stay, increasing their statistical risk profile.
- **Kennel Location Analysis:** We identified that the initial Kennel assignment (e.g., Medical Ward vs. General Population) is a strong proxy for the animal's health status at intake, serving as a critical predictor in our final boosting models.
- **Dimensionality Reduction (MCA):** To facilitate the visualization of our categorical clusters, we utilized **Multiple Correspondence Analysis (MCA)**. This technique allowed us to reduce the dimensionality of high-cardinality categorical data, projecting complex feature interactions into a lower-dimensional space to identify and display distinct risk archetypes clearly





Section 3: Methodology

3.1 Comparative Modeling Approach (Model Showdown)

Rather than selecting a single algorithm at the outset, we adopted an evolutionary benchmarking strategy. We evaluated five distinct architectures to identify the most effective "brain" for our prediction task:

1. **Baseline Model (Logistic Regression):** Established a linear statistical benchmark.
2. **Ensemble Model (Random Forest):** Tested bagging-based robustness and its ability to handle non-linear relationships.
3. **Advanced Boosting Suite (MLP, CatBoost, LightGBM):** Evaluated state-of-the-art gradient boosting machines specifically designed for tabular data.

Final Selection Rationale: After extensive testing, **LightGBM** was selected as our primary model. While **CatBoost** showed comparable precision, LightGBM achieved the highest **PR AUC (0.52) and Recall (0.81)**.

- **Metric Rationale:** A strategic decision was made to optimize for **PR AUC** rather than a single-point metric like Recall. While Recall is vital for animal safety, optimizing for it in isolation often forces the model to "cast too wide a net." PR AUC ensures the model is fundamentally better at distinguishing the minority class effectively across the entire precision-recall spectrum.
- **Leaf-wise Growth Strategy:** Unlike traditional level-wise trees (Random Forest), LightGBM grows trees by choosing leaves with maximum loss reduction. This allows it to capture complex, high-stakes patterns (like the intersection of specific breeds and intake types) more effectively.
- **Native Categorical Support:** This was a decisive factor. By treating features like Breed and Kennel as categorical objects rather than using One-Hot Encoding, we avoided high-dimensionality issues and preserved the structural information within the data.

3.2 Hyperparameter Optimization

To ensure the models reached their peak performance, we implemented **RandomizedSearchCV**.

- **Rationale:** Unlike a manual Grid Search, which is computationally expensive and limited, Randomized Search allowed us to efficiently explore a broad range of parameters (Learning Rate, Max Depth, Num Leaves). This strategic choice was critical in fine-tuning the balance between sensitivity and precision, ultimately maximizing the PR AUC score.

3.3 Strategy for Imbalanced Data: Cost-Sensitive Learning

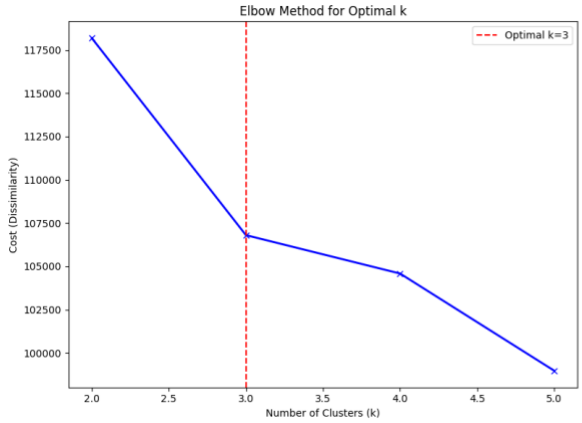
The "At-Risk" class represents only ~7% of the data. A standard model would optimize for accuracy by ignoring this minority class. To align the model with our ethical objective of "saving lives," we employed **Cost-Sensitive Learning**:

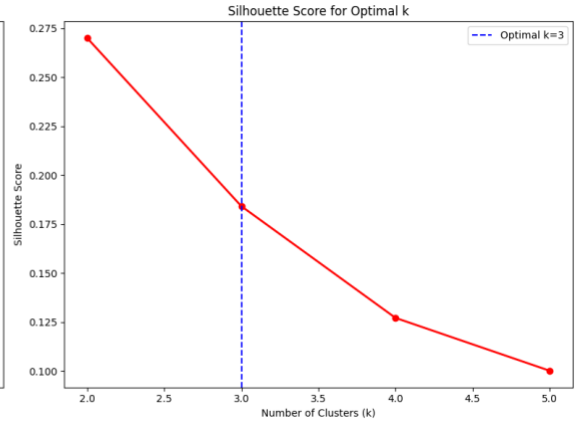
- **Class Weighting:** We utilized the is_unbalance=True (and scale_pos_weight) parameters. By assigning a higher penalty to "False Negatives" (missing an at-risk animal), we forced the model to prioritize the minority class.
- **Analytical Shift:** This choice shifted our focus from global accuracy to **PR AUC**. This ensures that the model provides high-quality predictions across the entire precision-recall spectrum, rather than just hitting a single point of Recall. By maximizing PR AUC, we ensure the model is robust in distinguishing 'At-Risk' animals despite the heavy class imbalance.

3.4 Unsupervised Analysis: Comparative Clustering & Risk Profiling

To validate our classification findings and uncover latent "Risk Archetypes," we implemented a comparative unsupervised learning framework.

- **Selection Rationale (K-Modes vs. K-Means):** Standard K-Means relies on Euclidean distance, which is mathematically invalid for categorical data (e.g., one cannot calculate the "mean" of discrete labels like "Pitbull" and "Lab"). Therefore, we primarily selected **K-Modes**, which utilizes a matching dissimilarity measure and updates centroids based on the **mode** (the most frequent category). This ensures the clusters are representative of real-world animal profiles rather than mathematical abstractions.
- **Algorithm Benchmarking (Hierarchical vs. K-Modes):** To ensure the robustness of our profiles, we conducted a comparative analysis between K-Modes and **Hierarchical Clustering**. We utilized the **Silhouette Score**—a metric that measures how similar an object is to its own cluster compared to others—as the primary tool for comparison. K-Modes achieved a superior Silhouette Score of **0.159** and better cohesion (WSS), confirming its effectiveness in identifying meaningful patterns within this categorical dataset.





Section 4: Experiments and Results

4.1 Experimental Design

- **Validation Scheme:** We employed a **Stratified Train-Test split (80/20)**. This was a strategic choice to ensure that the minority "Risk" class (~7%) maintains its proportional representation in both sets, preventing biased evaluation.
- **Algorithm Benchmarking:** Unlike a single-model approach, we tested five architectures: **Logistic Regression, Random Forest, MLP, CatBoost, and LightGBM**. This allowed us to hypothesize how different mathematical approaches (e.g., Bagging vs. Boosting) handle the high-cardinality categorical features of the shelter data.
- **Hyperparameter Tuning:** We utilized **RandomizedSearchCV** to optimize the winning LightGBM model. We focused on `learning_rate`, `num_leaves`, and `scale_pos_weight`. **Rationale:** Randomized search was chosen over Grid Search for its computational efficiency, allowing us to explore a wider parameter space to optimize for PR AUC performance.

4.2 Classification Performance

The optimized **LightGBM** model outperformed all other candidates, particularly in our primary objective of sensitivity.

Model Comparison Table:

	Model	PR AUC	Recall	Precision	F1-Score
0	LightGBM	0.522190	0.805556	0.286939	0.423152
1	CatBoost	0.513241	0.850000	0.253311	0.390306
2	Random Forest	0.446228	0.744444	0.275342	0.402000
3	MLP	0.429660	0.479630	0.405956	0.439728
4	Logistic Regression	0.348928	0.616667	0.260156	0.365934

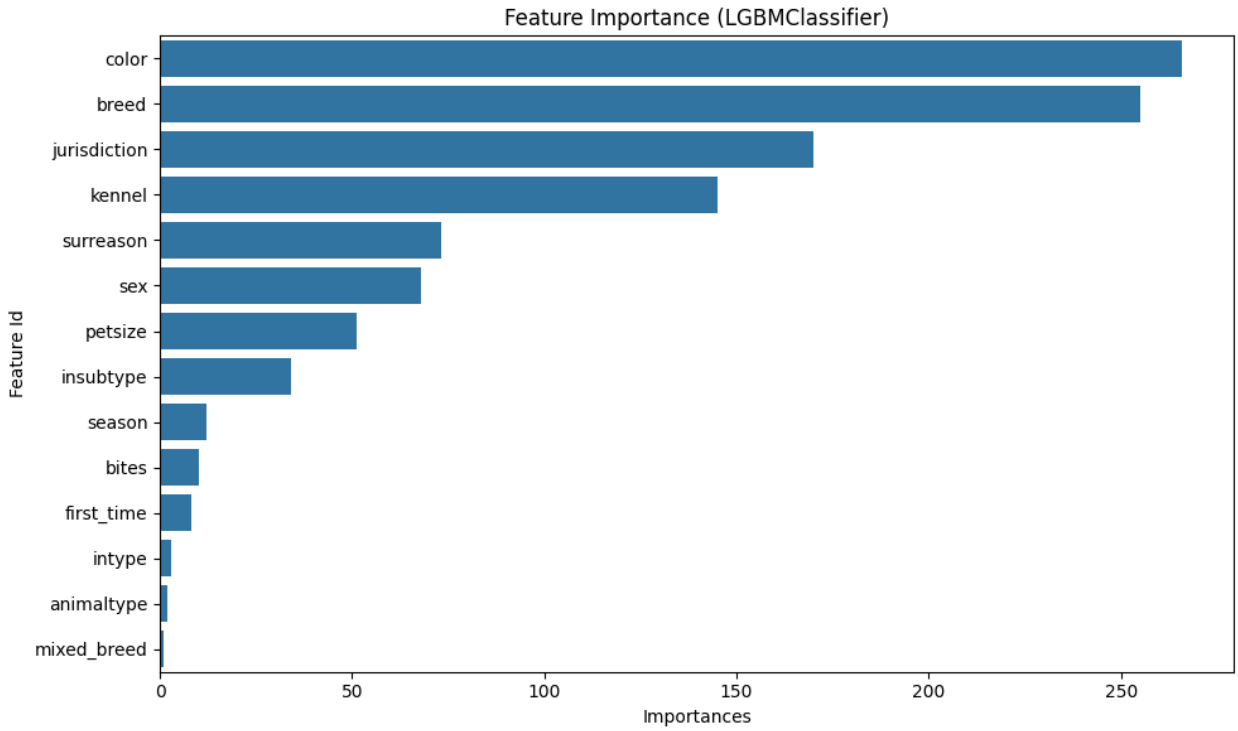
The model achieved a **PR AUC of 0.52**, confirming its ability to prioritize high-risk animals far more effectively than a random baseline (0.15). Our optimization strategy ensures that the risk scores generated are statistically significant across various threshold levels.

4.3 Feature Importance Analysis

The model identified several key drivers of risk, providing a distinction between technical predictors and actionable operational insights:

- **Technical vs. Operational Predictors:** While **Color** and **Breed** are identified as the top technical predictors due to their high descriptive diversity and cardinality in the dataset , **Kennel (Location)** and **Pet Size** emerge as the central operational indicators that allow shelter staff to perform immediate interventions.
- **Kennel (Location):** This remains a critical operational driver. **Rationale:** Initial placement in specialized areas like medical or isolation wards serves as a strong proxy for an animal's immediate health and behavioral status at intake.
- **Pet Size & Age:** The **PUPPY/KITTN** category emerged as a significant risk factor. **Rationale:** This aligns with the "Kitten Season" phenomenon where very young animals enter the shelter with high medical vulnerability during peak intake periods.
- **Intake Type:** Animals brought in as **"Strays"** or **"Confiscated"** showed higher risk profiles compared to owner surrenders.

Consistency with Domain Knowledge: The dominance of **Kennel** and **Pet Size** confirms that the initial triage-deciding where to house the animal and assessing its life stage-is the most indicative factor of its eventual outcome, shifting shelter management from reactive to proactive care.

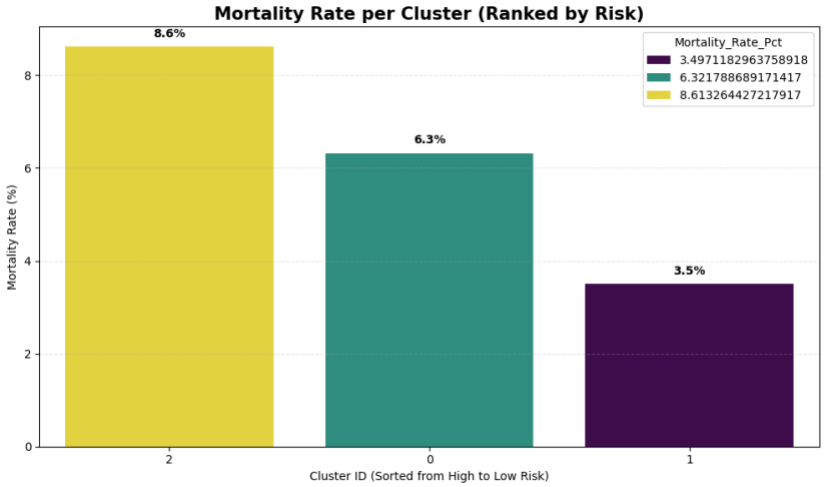


4.4 Unsupervised Risk Profiling (K-Modes vs. Hierarchical)

The unsupervised analysis was enhanced by comparing **K-Modes** and **Hierarchical Clustering** to identify the most stable "Risk Archetypes".

- **Model Selection & Validation**: K-Modes was selected as the superior algorithm for this categorical dataset, achieving a **Silhouette Score of 0.159**. This outperformed the hierarchical approach (Silhouette Score: 0.153) and demonstrated better internal cohesion with a lower Within-Cluster Sum of Squares (WSS).
- **Optimal Clustering**: Following the validation process, we partitioned the data into **3 distinct clusters**.
- **Findings from the High-Risk Cluster (Cluster 2)**:
    - **Mortality Rate**: **8.6%** (This represents the highest risk profile in the dataset, significantly higher than the average mortality rate).
    - **Centroid Profile**: Stray, Domestic Shorthair Cat, Intact Male/Female, admitted during **Summer**(July/August).
    - **Key Characteristics**: Black color, **PUPPY/KITTN** size category, and Intake Reason: **Stray**.

Interpretation: This segmentation confirms that risk is concentrated in a "perfect storm" of features. The profile of **Stray Cats in Summer** aligns with the peak of **"Kitten Season,"** where surges in intake lead to resource strain and higher medical vulnerability. Furthermore, the presence of **"Black"** as a dominant color in the high-risk centroid provides empirical evidence for **"Black Cat Bias"** (lower adoption rates), validating the model's ability to capture real-world sociodemographic challenges in shelter management.



Section 5: Conclusion and Discussion

5.1 Summary of Contributions

This project demonstrates a robust, data-driven framework for addressing high-stakes classification in animal welfare. Our primary contributions include:

- **Benchmarked Predictive Architecture:** We engineered a comparative pipeline across five diverse models. By optimizing for **PR AUC (0.52)**, our final **LightGBM** model provides a high-fidelity risk score. This ensures that the animals prioritized by the system are truly the most vulnerable, creating a dependable and balanced decision-support tool for shelter operations.
- **Strategic Feature Discovery:** Our analysis isolated **Initial Kennel Location** and **Pet Size/Age** as the primary drivers of animal risk. This allows shelters to shift from subjective intuition to empirical prioritization, targeting life-saving resources where they are most critically needed.
- **Scientifically Validated Risk Profiling:** Beyond classification, we implemented an unsupervised framework that was scientifically validated through a comparative analysis between **K-Modes** and **Hierarchical Clustering**. By achieving a **Silhouette Score of 0.159**, we provided empirical proof that the identified "Risk Archetypes" possess high internal cohesion rather than being random distributions. This analysis confirmed that risk is concentrated in specific profiles—such as stray cats entering during the peak of "Kitten Season"—providing actionable intelligence for proactive seasonal planning.

5.2 Team Roles

- **Kfir Diamond:** Responsible for the modeling and evaluation phase of the project. This included the comparative benchmarking of five machine learning architectures and performing hyperparameter optimization using RandomizedSearchCV. Also initiated the unsupervised analysis phase by implementing the K-Modes clustering algorithm to identify latent risk profiles.

- **Daniel Tugendhaft:** Led the comprehensive Data Engineering and preprocessing pipeline, including the definition of the binary target variable. Additionally, led the advanced unsupervised analysis by performing Hierarchical Clustering, validating cluster cohesion through the Silhouette Score, and utilizing Multiple Correspondence Analysis (MCA) to visualize high-dimensional categorical risk archetypes.

5.3 Limitations and Future Work

While the current model provides high sensitivity, several avenues for future research remain:

- **Natural Language Processing (NLP):** The dataset contains unstructured medical notes that currently go unutilized. Future iterations could employ NLP techniques to extract clinical keywords or behavioral triggers that are not captured in the structured categorical fields.
- **Advanced Text Embeddings:** The "Breed" field remains inherently noisy. Using pre-trained text embeddings (like Word2Vec or BERT) could help the model better understand the nuances and similarities between various breed descriptions, potentially improving Precision without sacrificing Recall.
- **Real-Time Integration:** A natural next step would be the deployment of the model via a simple API. This would allow shelter staff to receive an instantaneous "Risk Score" during the intake process, facilitating immediate decision-support for medical and behavioral triaging.